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Methodological Review

Performance of large language models in data extraction for evidence synthesis: A systematic review

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ABSTRACT

Objective: Data extraction is among the most resource-intensive and error-prone stages of systematic review production. Large language models (LLMs) offer potential for automating or semi-automating this process, yet their performance characteristics remain incompletely characterised. This systematic review aimed to comprehensively evaluate LLM accuracy, reliability, and efficiency for data extraction in evidence synthesis, and to identify optimal implementation strategies.

Methods: We searched PubMed, Embase, Web of Science, and preprint servers (medRxiv, arXiv) through December 2025. Studies were eligible if they evaluated one or more LLMs for data extraction against a human reference standard and reported quantitative performance metrics. Two reviewers independently extracted data and assessed methodological quality using PROBAST + AI and reporting completeness using TRIPOD-LLM. Narrative synthesis was performed due to substantial heterogeneity precluding *meta*-analysis.

Results: Twenty-seven studies met inclusion criteria, evaluating models including GPT-4/4o (n = 15), Claude versions 2–3.5 (n = 10), Gemini (n = 3), and open-source alternatives including Llama, Mistral, Qwen, and DeepSeek. Overall accuracy ranged from 47% to 99.9%, with substantial heterogeneity by task type and data granularity. Categorical and string variables were extracted more reliably (74–96%) than numerical data (47–88%). Claude 3.5 Sonnet achieved high accuracy in an assistive workflow (91.0%; 95% CI: 90.4–91.6%), exceeding human-only extraction (89.0%). Claude models outperformed GPT in head-to-head comparisons (OR 1.70 for event counts). Omissions were the dominant error type (60–74%), with hallucination rates of only 0.08–6%, challenging widespread fabrication concerns. Time savings of 33% to 87% were reported, although most included studies did not quantitatively assess efficiency. Methodological quality was generally robust, with 74.1% of studies rated low risk of bias under PROBAST + AI. Mean TRIPOD-LLM compliance was 88.5%, though gaps in inference settings and model version documentation were common.

Conclusion: LLMs demonstrate promising but variable performance for data extraction in evidence synthesis. Current evidence supports their integration as assistive tools within dual-extraction workflows requiring human verification, rather than as autonomous extractors. Categorical data is extracted more reliably than numerical outcomes, and few-shot prompting with structured output formats consistently improves performance. Standardised benchmarks and prospective comparative studies remain priorities for future research.

Introduction

Systematic reviews and *meta*-analyses represent the cornerstone of evidence-based practice, providing rigorous synthesis of available research to inform clinical and policy decisions [1,2]. However, the production of high-quality systematic reviews is resource-intensive, with study selection, data extraction, and critical appraisal representing the most time-consuming stages of the review process [1]. Traditional manual extraction requires trained reviewers to independently extract predetermined variables from primary studies, with discrepancies resolved through consensus, a process that can take hundreds of hours for comprehensive reviews. Data extraction is particularly

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problematic; error rates in manual extraction can reach up to 50%, resulting from the diverse backgrounds and varying statistical and clinical expertise of data abstractors [3]. Additionally, the velocity of research output continues to increase, with the International Prospective Register of Systematic Reviews (PROSPERO) registering over 100,000 systematic reviews by the end of 2020, compared to just 284 in 2011 [4].

The emergence of large language models (LLMs) has generated considerable interest in their potential to automate or semi-automate systematic review workflows [5]. Models such as OpenAI's GPT series, Anthropic's Claude, and Google's Gemini demonstrate remarkable capabilities in natural language understanding, text generation, and information extraction tasks that suggest potential applications in extracting structured data from unstructured scientific literature [6,7]. A recent systematic review identified 172 studies addressing LLM applications in review automation, with data extraction representing 31.4% of the focus areas, second only to literature searching [5]. When comparing pooled performance metrics, GPT-based models achieved mean precision of 83.0% and recall of 86.0% for data extraction tasks, outperforming traditional BERT-based approaches [5].

Several methodological developments have accompanied this rapid adoption. The National Institute for Health and Care Excellence (NICE) published a position statement in August 2024 outlining expectations for the use of artificial intelligence methods in evidence generation, emphasizing that AI should enhance rather than replace human involvement [8]. The statement explicitly addresses systematic review automation, recommending adherence to established guidelines such as PALISADE for machine learning applications and the newly developed TRIPOD-LLM for studies using large language models [9]. TRIPOD-LLM provides a comprehensive 19-item checklist with 50 subitems specifically addressing the unique challenges of LLMs in biomedical applications, including concerns regarding hallucinations, omissions, reproducibility, and biases [9]. Furthermore, Cochrane has begun integrating AI technologies into its review workflows, signaling mainstream acceptance of these approaches within evidence synthesis [10].

Collaborative LLM workflows represent an emerging paradigm in this field. Khan et al. [3] demonstrated that simulating a two-reviewer process using GPT-4-turbo and Claude-3-Opus achieved 94% accuracy for concordant responses, suggesting that multi-model architectures may enhance reliability. Such approaches align with NICE's emphasis on maintaining human oversight while leveraging AI efficiency gains. Similarly, validation studies have shown that LLM-assisted extraction can reduce extraction time by 33% to 83% while maintaining or exceeding human accuracy rates [11,12].

Despite these promising developments, no comprehensive synthesis of the rapidly evolving literature on LLM performance for data extraction exists. Understanding the current state of evidence regarding LLM accuracy across different data types, optimal prompting strategies, efficiency gains, and limitations is essential for researchers considering integration of these tools into their workflows. This is particularly timely given regulatory guidance emphasizing transparent reporting and validation requirements.

This systematic review aims to comprehensively evaluate the performance of LLMs in data extraction for evidence synthesis. Specifically, we sought to: (1) characterize the accuracy and reliability of LLM-based extraction across different data types and clinical domains; (2) identify optimal prompting strategies and methodological approaches; (3) quantify efficiency gains relative to manual extraction; and (4) delineate current limitations and requirements for human oversight.

Statement of Significance

Problem or Issue	Data extraction in systematic reviews is time-intensive and error-prone, yet no comprehensive synthesis of LLM performance for this task exists to guide researchers.

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Problem or Issue	Data extraction in systematic reviews is time-intensive and error-prone, yet no comprehensive synthesis of LLM performance for this task exists to guide researchers.
What is Already Known	LLMs demonstrate natural language processing capabilities applicable to evidence synthesis, and individual studies report variable extraction accuracy. However, findings are fragmented across heterogeneous designs with no unified quality assessment.
What this Paper Adds	This is the first systematic review applying both PROBAST + AI and TRIPOD-LLM frameworks to 27 studies, establishing that LLM accuracy ranges from 47 to 99.9%, omissions dominate over hallucinations (60–74% vs. 0.08–6%), and Claude models outperform GPT in direct comparisons. Time savings of 33% to 87% are achievable with human-in-the-loop workflows.
Who Would Benefit	Systematic review methodologists, evidence synthesis researchers, clinical informaticians, and journal editors evaluating AI-assisted review workflows.

Methods

Search strategy and eligibility criteria

We conducted a comprehensive search of PubMed, Embase, Web of Science, and preprint servers (medRxiv, arXiv) from database inception to 15 December 2025. Search terms combined concepts related to large language models (including specific model names), systematic reviews/evidence synthesis, and data extraction. Reference lists of included studies were screened for additional eligible studies. The review was conducted and reported in accordance with the PRISMA 2020 statement, and the completed PRISMA 2020 checklist is provided as Supplementary Appendix S1. The full, database-specific search strategies—including all keywords, Boolean operators, MeSH and Emtree terms, field tags, and any database-specific adaptations—are provided in Supplementary Appendix S2 to ensure reproducibility. In brief, the representative PubMed strategy combined three concept blocks with the Boolean operator AND: (i) large language models ("large language model" OR "large language models" OR "language model" OR "language models" OR LLM OR LLMs OR GPT* OR ChatGPT OR "generative AI" OR "generative artificial intelligence" OR "foundation model" OR "foundation models" OR "generative model" OR "generative models" OR chatbot OR chatbots OR "prompt engineering" OR "Natural Language Processing" OR "Artificial Intelligence" OR "Deep Learning"); (ii) evidence synthesis ("systematic review" OR "systematic reviews" OR *meta*-analys* OR "evidence synthesis" OR "evidence syntheses" OR "systematic literature review" OR "systematic literature reviews" OR "scoping review" OR "scoping reviews" OR "literature review" OR "literature reviews" OR "rapid review" OR "rapid reviews" OR "umbrella review" OR "umbrella reviews" OR "living systematic review" OR "knowledge synthesis" OR "research synthesis" OR Cochrane OR PRISMA OR "Systematic Reviews as Topic" OR "Meta-Analysis as Topic" OR "Review Literature as Topic"); and (iii) data extraction ("extraction" OR "data extraction" OR "data abstraction" OR "information extraction" OR "data charting" OR "evidence extraction" OR "extracting data" OR "extract data" OR "data extracted" OR "automated extraction" OR "automatic extraction" OR "data collection" OR "outcome extraction" OR "study characteristics" OR PICO OR PICOS OR "table extraction" OR "structured extraction"), with within-block terms combined using OR. The review protocol was prospectively registered with PROSPERO (CRD420261431446); the eligibility criteria, outcomes, and data-extraction framework were specified a priori and adhered to throughout, with no substantive deviations from the registered protocol.

We structured the review around a methodological PICO-style framework, and organised the eligibility criteria, extraction variables, and synthesis around these predefined categories. The Population/context comprised primary studies (or their abstracts, tables, and

figures) undergoing data extraction for evidence synthesis. The Intervention comprised LLM-based extraction, further categorised as (a) base/foundation models accessed directly, (b) LLM-powered commercial tools or wrappers, and (c) human-in-the-loop (LLM-assisted) workflows. The Comparator/reference standard was human extraction (single- or dual-reviewer). The Outcomes were extraction accuracy and reliability, error type (omission versus hallucination), and efficiency. Accordingly, the primary review question was: among primary studies undergoing data extraction for evidence synthesis, how accurate, reliable, and efficient is LLM-based extraction relative to a human reference standard, and how does performance vary by tool type, data type, and workflow? Studies were eligible if they: (1) evaluated one or more LLMs for data extraction tasks; (2) applied extraction to primary studies intended for systematic review, *meta*-analysis, or other evidence synthesis; (3) compared LLM performance against a human reference standard; and (4) reported quantitative performance metrics. We excluded studies focused solely on screening/selection tasks, risk of bias assessment without data extraction, or non-systematic literature reviews. Title/abstract screening and full-text review were each performed independently by two reviewers, with conflicts resolved by discussion and, where consensus could not be reached, adjudication by a third reviewer; all screening was managed in Covidence. Preprints (medRxiv, arXiv) were eligible and were cross-checked against the published literature so that any subsequently peer-reviewed version of the same study was retained only once; overlapping or duplicate reports of a single study were linked to one study record to avoid double-counting.

Data extraction and quality assessment

Two reviewers independently extracted data using a standardized form. Extracted variables included study characteristics (design, domain, sample size), LLM specifications (model, version, access method, fine-tuning status), task characteristics (extraction type, variable complexity), prompting strategies (zero-shot, few-shot, iterative refinement), reference standard details, and performance outcomes (accuracy, precision, recall, F1, agreement measures, error rates). Efficiency outcomes (time savings, costs) were extracted where reported.

Methodological quality was assessed using PROBAST + AI, a validated tool for evaluating prediction models incorporating artificial intelligence. Risk of bias was evaluated across four domains: participants and data sources, predictors, outcome, and analysis. Each domain was rated as low, high, or unclear risk. Reporting completeness was assessed using TRIPOD-LLM, a reporting guideline specifically developed for studies using large language models [9]. Two reviewers independently assessed all studies, with disagreements resolved through discussion. Due to heterogeneity in study designs, populations, and outcomes, *meta*-analysis was not performed; findings are synthesized narratively.

Results

We conducted a comprehensive search of four electronic databases from inception to December 2025: Embase (n = 2,145), PubMed (n = 1,876), Web of Science (n = 1,634), and preprint servers (medRxiv, arXiv) (n = 342), yielding 5,997 records. Search terms combined concepts related to large language models (including specific model names), systematic reviews/evidence synthesis, and data extraction. Additional references were identified through citation searching (n = 18). The systematic review process was managed using Covidence systematic review software. Following removal of 2,287 duplicates (35 manually, 2,252 by Covidence), 3,728 records were screened by title and abstract, with 3,165 excluded as not meeting the predefined eligibility criteria (most commonly because the record did not evaluate LLM-based data extraction, lacked a human reference standard, or reported no quantitative performance metric). Of the 563 studies assessed for full-text eligibility, 536 were excluded for wrong population, setting, or phenomenon (n = 148), protocols or publication type issues (n = 42), wrong study design (n = 338), or methodological limitations (n = 8). Twenty-seven studies met inclusion criteria and were included in the review (Fig. 1).

Study selection and characteristics

Twenty-seven studies met inclusion criteria and were included in the review. Table 1 summarizes the characteristics of included studies. Studies originated from 13 countries, with the United States (n = 8) and China (n = 4) contributing the most publications, followed by Denmark, Germany, Italy, and the United Kingdom (n = 2 each). Most studies used validation (n = 8) or comparative validation (n = 8) designs, with proof-of-concept designs in six studies and other designs (prospective study within reviews, pre-registered validation, exploratory, development and usability, and rapid feasibility) in the remaining five.

Clinical domains were diverse, including psychology [13], oncology [3], dentistry [14,15], sleep medicine [16], dermatology [17], geriatric medicine [18], and multiple medical specialties [11,19]. Non-clinical domains included software engineering [20], educational technology [21,22], and business process management [23].

The number of primary studies processed ranged from 8 [24] to 2,179 [13], with most studies evaluating extraction from 10 to 100 articles. Data sources were predominantly full-text PDF articles, though some studies used abstracts only [7] or registry data in combination with publications [25].

Large language models evaluated

Studies evaluated a range of commercial and open-source LLMs. OpenAI models were most frequently assessed, including GPT-4 and GPT-4-turbo (n = 6), GPT-4o and GPT-4o-mini (n = 9), GPT-4.1 (n = 1; (24)), GPT-3.5 (n = 4), and the reasoning-enhanced o3 model (n = 1; (25)), and GPT-5 (n = 1; (31)). Anthropic's Claude models were evaluated in 10 studies, spanning versions 2/2.1 [11,17,19,32], 3 Opus (3), 3/3.0 [11,13], and 3.5 Sonnet [11,12,14,16,30].

Google's Gemini models (1.5 Pro, 1.5 Flash, 2.0-flash) were evaluated in three studies [14,21,23]. Open-source models included Llama 3.1/3.3 [13,23], Mistral Large/Medium [13,21], Qwen 2.5 [23,31], and DeepSeek v3 R1 [14]. Specialized tools built on LLMs, including Elicit [27,34], ChatPDF [29,32], and MetaMate [22], were also evaluated.

Access methods varied between API access (n = 13) and web interface (n = 14). All but one study [28], using custom GPT configuration, employed off-the-shelf models without fine-tuning. Throughout this review we distinguish base/foundation models (e.g., GPT-4/4o, Claude, Gemini, Llama, Mistral, Qwen, DeepSeek), evaluated directly via API or web interface, from LLM-powered tools or platforms that wrap an underlying model behind a proprietary interface (e.g., Elicit, ChatPDF, Jenni AI, Microsoft Bing AI, MetaMate, and custom GPTs). For the latter, reported performance reflects the combined effect of the base model, the tool's retrieval and prompting layer, and its output formatting, and is therefore not directly attributable to the base model alone. This base-model versus tool distinction is reflected in the model labelling in Tables 1 and 3 and should be borne in mind when interpreting cross-study comparisons.

Data extraction tasks and prompting strategies

Table 2 summarizes extraction task characteristics and prompting approaches. Extraction tasks encompassed study characteristics (author, year, design, setting), PICO elements (population, intervention, comparator, outcomes), numerical data for *meta*-analysis (sample sizes, event counts, means, standard deviations), risk of bias items, and complex subjective assessments (e.g., SWOT analysis; [30].

Variable types included categorical/text data, continuous numerical values, and binary outcomes. The number of variables extracted per study ranged from 4 (structured fields per dataset; [33]) to 67

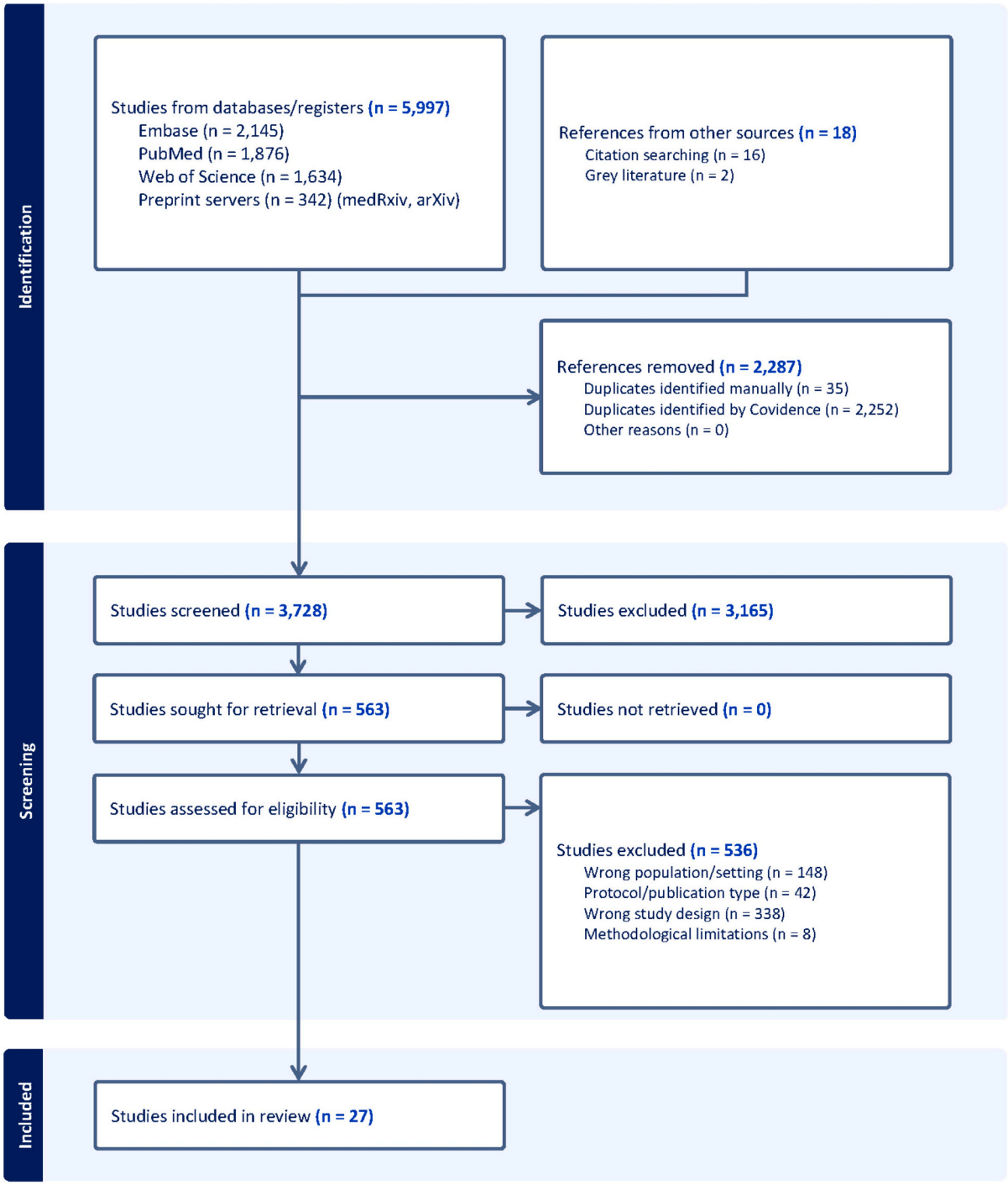


Fig. 1. PRISMA Flow Diagram.

(comprehensive extraction; [25]; Liu et al. [19] extracted 58 items across six domains). Several studies explicitly categorized variables by complexity, distinguishing "explicitly stated" data from "derived" variables requiring inference [21,30].

Prompting strategies varied considerably. Zero-shot approaches were used in 8 studies, while few-shot prompting (1–6 examples) was employed in 9 studies. Several studies used structured output formats including JSON schemas [14,20,23] or predefined table templates [16,19]. Iterative prompt refinement was reported in 15 studies. Schmidt et al. (2024) [7] notably required approximately 20 iterations for optimal prompt development.

Performance outcomes

Overall accuracy

Table 3 presents detailed performance outcomes across studies. Overall accuracy ranged widely from 47% (Jansen et al., 2025 [13], for overall extraction including effect sizes) to 99.92% (Caponio et al., 2025 [14], for sub-outcome level single data points). Most studies reported accuracy in the 65–90% range for mixed extraction tasks.

Gartlehner et al. (2025) [11], in the largest and most methodologically rigorous study (63 studies across 6 systematic reviews, 9,341 data elements), reported overall accuracy of 91.0% (95% CI: 90.4–91.6%) for

Table 1
Characteristics of included studies.

First Author, Year	Country	Study Design	Clinical Domain	Review Type	No. Studies	Data Source	LLM Model(s)	Access	Fine-tuning
Peng, 2025 (16)	China	Proof-of-concept	Sleep medicine	MA/SR	648 RCTs	Full-text PDF	ChatGPT-4o, Claude 3.5	Web	No
Caponio, 2025 (14)	Italy	Comparative validation	Dentistry	MA	60 articles	Full-text PDF	GPT-4o, Claude 3.5, DeepSeek, Gemini	API	No
Ronkainen, 2025 (24)	Finland	Proof-of-concept	Ageing biomarkers	SR/MA	8 articles	Full-text PDF	GPT-3.5, GPT-4.1	API	No
Liu, 2025 (19)	China	Validation	Multiple medical	SR (Cochrane)	10 RCTs	Full-text PDF	Claude-2	Web	No
Daraqel, 2025 (15)	Palestine	Comparative validation	Orthodontics	SR	300 studies	Full-text PDF	Microsoft Bing AI	Web	No
Gartlehner, 2025 (11)	USA	Prospective SWAR	Multiple medical	SR	63 studies	Full-text PDF	Claude 2.1, 3.0, 3.5	Web	No
Kataoka, 2025 (25)	Japan	Validation	Insomnia; Psoriasis	SR/NMA	305 RCTs	PDF, tables, figures	GPT-4o, o3	API	No
Khraisha, 2024 (6)	Ireland	Pre-registered validation	Refugee parenting	SR	30 full texts	Full-text, abstracts	GPT-4	Web	No
Jensen, 2025 (18)	Denmark	Comparative validation	Geriatric/falls	SR	11 RCTs	Full-text PDF	ChatGPT-4o	Web	No
Khan, 2025 (3)	USA	Validation	Oncology (prostate)	Living SR/ NMA	22 publications	PDF (text, tables)	GPT-4-turbo, Claude-3-Opus	API	No
Mahmoudi, 2025 (26)	USA	Exploratory	COVID-19 modeling	SLR	10 studies	Full-text PDF	ChatGPT (GPT-4)	Web	No
Jansen, 2025 (13)	Germany	Validation	Psychology	MA of SRs	2,179 studies	Full-text PDF	GPT-4o, Claude 3, Llama, Mistral	API	No
Gartlehner, 2024 (17)	USA	Proof-of-concept	Dermatology	SR	10 RCTs	Full-text PDF	Claude 2	Web	No
Felizardo, 2024 (20)	Brazil	Proof-of-concept	Software Engineering	SMS	25 studies	Full-text PDF	ChatGPT-4.0 (GPT-4o)	API	No
Bianchi, 2025 (27)	Switzerland	Comparative	Healthcare/Nursing	Validation	20 RCTs	Full-text PDF	Elicit (GPT-3)	Web	No
Sercombe, 2025 (28)	Australia	Development	Mental health	SR	20 studies	Full-text PDF	Custom GPT (GPT-3.5)	Web	Custom
Meliante, 2025 (29)	Italy	Comparative	Ophthalmology	SLR	106 studies	Full-text PDF	Elicit, ChatPDF, Jenni AI	Web	No
Stewart-Evans, 2025 (30)	UK	Proof-of-concept	Spatial planning/ health	Scoping	10 sources	Full-text PDF	Claude 3.5 Sonnet	Web	No
Schmidt, 2024 (7)	UK	Rapid feasibility	Multiple domains	SR	30 studies	Abstracts only	GPT-4	API	No
Long, 2025 (31)	USA	Validation	Health professions ed.	Scoping	187 manuscripts	Full-text	GPT-4o, GPT-4o-mini, GPT-5, Qwen 2.5	API	No
Sun, 2024 (32)	China	Comparative	Neuropsychiatry; Sleep	SR/MA	88 RCTs	Full-text PDF	ChatPDF (GPT-3.5), Claude 2	Web	No
Lai, 2025 (12)	China	Validation	Complementary medicine	SR (Cochrane)	107 RCTs	Full-text PDF	Moonshot-v1-128 k (Kimi), Claude 3.5 Sonnet	API	No
Schroeder, 2025 (21)	USA	Comparative validation	Educational technology	Scoping	112 studies	Full-text PDF	Gemini 1.5 Pro/ Flash, Mistral Large 2	API	No
Ateia, 2025 (23)	Germany	Proof-of-concept	Business Process Mgmt	Lit review	122 papers	Full-text PDF	Qwen 2.5, Llama 3.3, Gemini	API	No
Wang,2024 (22)	USA	Validation	Educational research	MA/SR	32 studies	Full-text	GPT-4	API	No
Wang, 2025 (33)	USA	Validation	Nutrition; NLP/ML	SR	40 papers	PDF (multimodal)	GPT-4o, GPT-4, GPT-3.5	API	No
Andersen, 2025 (34)	Denmark	Comparative	Diabetes; Cardiometabolic	SR	30 articles	Full-text PDF	Elicit, ChatGPT GPT-4o	Web	No

Abbreviations: MA, *meta*-analysis; SR, systematic review; NMA, network *meta*-analysis; SMS, systematic mapping study; SLR, systematic literature review; NR, not reported.

AI-assisted extraction using Claude 3.5 Sonnet, with precision of 98.9% and sensitivity of 89.4%. Notably, AI-assisted extraction outperformed human-only extraction (91.0% vs. 89.0%), suggesting potential for LLMs to reduce human error rates.

Agreement measures (Cohen's kappa) between LLM and human extraction ranged from 0.19 (slight; Schroeder et al., 2025 [21] with Mistral Large 2) to 0.96 (almost perfect; test–retest reliability for ChatPDF reported by Sun et al., 2024 [32]). Peng et al. (2025) [16] did not report pooled kappa values; reliability testing on two 5% random samples showed substantial agreement for 54.2% and 58.3% of items and almost perfect agreement for 45.8% and 41.7%.

Performance by variable type and complexity

A consistent finding across studies was superior performance for categorical/string variables compared to numerical data. Liu et al. (2025) [19] reported 94.77% overall accuracy, which fell to 77.97% for baseline characteristics, with 80.6% of all errors occurring in participant-number items. Kataoka et al. (2025) [25] found string variables achieved 74.0–81.7% accuracy compared to 60.4–74.2% for numeric variables. Schroeder et al. (2025) [21] demonstrated the same pattern, with explicitly stated data achieving 81–85% accuracy versus 65–70% for derived categorical variables.

Gartlehner et al. (2025) [11] stratified performance by data type:

Table 2
Data Extraction Tasks and Prompting Strategies.

Study	Extraction Task Type	Variable Types	No. Variables	Prompting Approach	Examples/ Schemas	Iterative Refinement	Reference Standard
Peng, 2025 (16)	Numerical MA data (group sizes, events, means, SDs)	Continuous, binary	5,714 data points	Single/multiple-sentence	Yes (table format)	Yes	Error-corrected DB; dual extraction
Caponio, 2025 (14)	Numerical MA outcomes	Continuous, binary	3,623 sub-outcomes	Zero-shot (SMART format)	Yes (JSON)	No	Dual independent reviewers
Ronkainen, 2025 (24)	Study characteristics (8 fields)	Text, categorical	25 fields (4 categories)	Few-shot (LangChain)	Yes (JSON)	Yes	Human extraction
Liu, 2025 (19)	Comprehensive (characteristics, PICO, outcomes)	Text, categorical, numerical	58 items	Zero-shot structured	Yes (table)	Yes	Cochrane review data
Daraqel, 2025 (15)	Study characteristics and outcomes	Mixed	15 per study	Not detailed	NR	NR	Dual reviewers; consensus
Gartlehner, 2025 (11)	Study characteristics (7–8 fields)	Text, categorical, numerical	9,341 total fields	Few-shot (4 studies)	Yes (questions)	Yes	Dual extraction; blinded
Kataoka, 2025 (25)	67 variables (comprehensive)	Text, categorical, numerical	67 variables	Step-by-step/ codebook/batch	Yes (codebook)	Yes	Dual extraction; consensus
Khraisha, 2024 (6)	Characteristics and eligibility	Text, categorical	NR	High-reliability prompts	NR	Yes	≥2 human reviewers
Jensen, 2025 (18)	Mixed (characteristics, PICO, outcomes)	Mixed	22 per study	Zero-shot natural language	NR	NR	Single human extraction
Khan, 2025 (3)	Granular MA data (PICO, outcomes, RoB)	Mixed	23 variables (506 responses)	Zero-shot + cross-critique	Yes (structured)	Yes	Two-LLM collaborative
Mahmoudi, 2025 (26)	Comprehensive characteristic	Mixed	6 per study	Structured prompts	Yes	Yes (7 iterations)	Dual human extraction
Jansen, 2025 (13)	TOPICS + M and effect sizes	Categorical, numerical	186 variables	Single-step codebook	Yes (codebook)	No	Published SR databases
Gartlehner, 2024 (17)	Characteristics, PICO, outcomes	Mixed	16 items	Conversational zero-shot	No	Yes	Consensus extraction
Felizardo, 2024 (20)	SMS characteristic	Text, categorical	10 items	Zero-shot	Yes (JSON)	Minimal	Expert verification
Bianchi, 2025 (27)	Characteristics, PICO, outcomes	Text, categorical	7 variables	Customizable commands	NR	NR	Dual human extraction
Sercombe, 2025 (28)	Characteristics, statistics	Mixed	Variable	RISEN framework	Yes	Yes	Adjudicated extraction
Meliante, 2025 (29)	Study composition	Text, categorical	NR	Not detailed	NR	NR	PRISMA-based SRs
Stewart-Evans, 2025 (30)	Citation details, SWOT analysis	Text, categorical	8 items	Simple/extended protocols	Yes	Yes	Human extraction
Schmidt, 2024 (7)	PICO elements; characteristic	Text, categorical	7–11 items	Zero-shot, JSON schema	Yes	Yes (~20 iterations)	Single human extraction; two reviewers compared
Long, 2025 (31)	Scoping review characteristic	Text, categorical	17 questions	Multi-agent scheme	Yes	Yes	Human extraction
Sun, 2024 (32)	Numerical MA data	Continuous, binary	4 (binary) or 6 (continuous) tasks per trial	Structured prompts	Yes (table)	Yes	Verified metadata
Lai, 2025 (12)	Comprehensive (characteristics, outcomes, RoB)	Mixed	~120 per RCT	Few-shot learning	Yes	Yes	Dual; Cochrane standards
Schroeder, 2025 (21)	Study characteristics, PICO-like elements	Text, categorical	24 variables	Zero-shot with schemas	Yes (schemas)	Minimal	Single expert coder
Ateia, 2025 (23)	Semantic concepts	Text, categorical	32 targets (15 categorical, 13 binary, 4 free-text)	JSON-based prompts	Yes (JSON)	Yes	Expert consensus
Wang,2024 (22)	Characteristics, effect sizes, moderators	Mixed	20 elements	Structured JSON output	Yes	Yes	Published MA data
Wang, 2025 (33)	Structured data from text/ tables/figures	Mixed	4 per dataset	Interactive standardization	Yes	Yes	Human validation
Andersen, 2025 (34)	SR characteristic	Mixed	180 data points	Structured prompts	Yes	NR	Human extraction

Abbreviations: MA, *meta*-analysis; PICO, population/intervention/comparator/outcome; RoB, risk of bias; NR, not reported; DB, database.

string data achieved 91.7% accuracy compared to 87.8% for numerical data. Sun et al. (2024) [32] demonstrated particularly stark differences, with binary outcomes achieving 83.3–85.9% accuracy versus only 47.8–56.1% for continuous outcomes.

Jansen et al. (2025) [13], in the largest study of effect size extraction (2,179 primary studies), reported notably lower performance: 32% accuracy for effect sizes overall, with 52% of values omitted entirely. Variable complexity also influenced performance. Stewart-Evans et al. (2025) [30] found 83–100% accuracy for simple citation items but only 10–16% for complex subjective assessments.

Head-to-head model comparisons

Several studies directly compared multiple LLMs. Peng et al. (2025) [16] found Claude 3.5 consistently outperformed ChatGPT-4o across all four extraction tasks, with the largest difference for event counts (12.7

Table 3
Performance outcomes by study.

Study	LLM Model(s) Tested	Overall Accuracy	Agreement (κ)	Error/Hallucination Rate	Performance by Data Type	Key Comparison Findings
Peng, 2025 (16)	Claude 3.5, ChatGPT-4o (single & multiple-sentence prompts)	Claude: 65.7–71.5%; ChatGPT: 52.9–69.1% (both prompt types)	Reliability samples: substantial (54.2%, 58.3%) to almost perfect (45.8%, 41.7%) agreement; pooled kappa not reported	NR	Group sizes best (71.5%); SDs lowest (59.9–68.9%)	Claude > ChatGPT (OR = 1.70, $p < 0.05$); Multiple-sentence > single-sentence
Caponio, 2025 (14)	ChatGPT-4o, Claude 3.5, DeepSeek v3 R1, Gemini 2.0-flash	Sub-outcome: 99.92%; Outcome: 95.83%; Study-level: 89.17%	Cochran's Q $p = 0.0047$	Hallucinations: 0.08% (3/3,623); Omissions: 60.9–74.5%	Single data points: near-perfect; Aggregated: higher errors	No significant model differences; Omissions ≫ hallucinations
Ronkainen, 2025 (24)	GPT-3.5-turbo, GPT-4.1	NR (recall-focused)	NR	NR	Metadata: reliable; Results: challenging	GPT-4.1: recall 0.66, F1 0.79 for results matched to gold standard; 41/122 gold-standard results retrieved; precision not reported
Liu, 2025 (19)	Claude-2	Overall: 94.77%; Baseline characteristics: 77.97% (lowest); Others: 100%	NR	98/1,873 errors (5.2%)	Baseline characteristics poorest (77.97%); 80.6% of errors in participant-number items	Single model; 80% errors in participant numbers
Daraqel, 2025 (15)	Microsoft Bing AI	Accuracy not reported in these terms	Kappa 0.16–1.00 (categorical); ICC 0.85–1.00 (continuous)	Analysed participants extracted from only 68% (204/300) of studies; 230 studies misclassified as "NR" for design type	Age/demographics: excellent; Complex: lower	AI extracted only 68% vs 99.4% for human (analyzed participants)
Gartlehner, 2025 (11)	Claude 3.5 Sonnet (AI-assisted workflow)	91.0% (95% CI: 90.4–91.6); P = 98.9%, R = 89.4%, F1 = 0.94	Concordance: 77.2%	Missed: 5.5%; Misallocated: 1.8%; Hallucinated: 0.8%	String: 91.7%; Numeric: 87.8%; Mixed: 85.4%	AI-assisted (91.0%) > human-only (89.0%); 41 min/study saved
Kataoka, 2025 (25)	GPT-4o, o3 (step-by-step, codebook, batch methods)	GPT-4o: 61.6%; o3: 75.3% (Dataset 2); Batch: 96.3% (Dataset 3)	Variable detection: 92.2–93.5%	Batch: hallucination concerns	String: 74.0–81.7%; Numeric: 60.4–74.2%	o3 > GPT-4o; Batch most efficient but needs verification
Khraisha, 2024 (6)	GPT-4	English peer-reviewed: 82%; Grey lit: 81%; Other languages: 85%	Adjusted κ = 0.62–0.69 (data extraction)	Hallucinations noted	Better on imbalanced datasets	High-reliability prompts essential; over-exclusion bias
Jensen, 2025 (18)	ChatGPT-4o	92.4% (224/242 correct)	Gwet's AC2: 0.93	False data: 5.2%; Missing: 2.3%	General info: 98.2%; Outcomes: 73.9%	Suitable as second rater; hallucinations when info unreported
Khan, 2025 (3)	GPT-4-turbo + Claude-3-Opus (collaborative)	Concordant responses (87%): 94%, F1 0.96; GPT-4-turbo: 89%; Claude-3-Opus: 90%	87% LLM agreement	GPT-4 T: 62% variable-level; Claude: 56% context-level errors	Trial characteristics: perfect; Complex variables: lower	Collaborative two-LLM > individual models; 94% for concordant
Mahmoudi, 2025 (26)	ChatGPT (GPT-4)	Overall: 43.3% (iteration 1) to 71.7% (final); Study settings: 93.3%; Behavioural components: 50%	NR	Fabrications/ overreporting noted	Location: 100%; Complex: max 80%	Iterative refinement essential; 7 iterations needed
Jansen, 2025 (13)	GPT-4o, GPT-4o mini, Claude 3, Llama 3.1, Mistral	TOPICS + M: 69%; Effect sizes: 32%; Overall: 47%	Mean: 63% (range 5–100%)	Hallucinations: 2–6%; Omissions: 52% (effect sizes)	Categorical: 74%; Effect sizes: 7–36%	GPT-4o best; Effect size extraction unreliable
Gartlehner, 2024 (17)	Claude 2	Overall: 96.3% (154/160)	Test-retest: 96.9% and 95.0% accuracy (kappa not reported)	4 missed items	Simple: high; Complex: lower	Test-retest: 95.0–96.9%; time savings not directly measured
Felizardo, 2024 (20)	ChatGPT-4.0 (GPT-4o)	Overall: 87.83%; Bibliometric: 98.6%; RQ-related: 65.11%	NR	FP (hallucinations): 22; FN: 8	Bibliometric: excellent; Complex RQ: lower	Precision/recall/F1 not reported (TP = 321, TN = 4, FP = 22, FN = 23)
Bianchi, 2025 (27)	Elicit (GPT-3 based)	Accuracy not reported; more than human 29.3%, equal 20.7%, partially equal 45.7%, deviating 4.3%	Custom classification; no kappa reported	NR	Study design: 100% "more"; Intervention effects: 95% "partially equal"	Wide variability; 95% partially equal for effects
Sercombe, 2025 (28)	Custom GPT (Systematic Review Extractor Pro)	Overall agreement: Review 1 91.45%; Review 2 89.31%	Subsections: 87.07–95.25% (R1); 77.50–90.83% (R2)	Hallucinations noted	Characteristics > statistical results	Statistical results poorest (77.50%); ~22 min saved per study
Meliante, 2025 (29)	Elicit, Connected Papers, ChatPDF, Jenni AI	Elicit: 51.4%; ChatPDF: 60.3%	NR	Incorrect: 12.5% (Elicit), 14.7% (ChatPDF)	Main text > tables/ figures	ChatPDF > Elicit; No single tool best for all
Stewart-Evans, 2025 (30)	Claude 3.5 Sonnet	Simple: 83–100%; Complex: 10–16%	Reliability: 0.55–0.94	Hallucinations: 1–2 per 10 sources	Citation items: 83–100%; Subjective SWOT: 10–16%	Best for simple extraction only

(*continued on next page*)

Table 3 (*continued*)

Study	LLM Model(s) Tested	Overall Accuracy	Agreement (κ)	Error/Hallucination Rate	Performance by Data Type	Key Comparison Findings
Schmidt, 2024 (7)	GPT-4	Human clinical: 82%; Animal: 80%; Human social science: 72% (overall 78% complete)	Not formally calculated; F1 0.76–0.78 (EBM-NLP)	31% string responses changed on re-run	Boolean/numerical: 90–100%; Free-text: 67–77%	High prompt sensitivity; stability concerns
Long, 2025 (31)	GPT-4o (multi-agent MAKMAO)	75.8% of responses concordant with human extraction; AI inaccuracies: 1.5%	Kappa 0.25–0.99 (categorical); thematic similarity 0.45–1.0 (free text)	AI inaccurate: 1.5%; Interpretation diff: 18.3%	Factual: high; Interpretive: variable	Multi-agent > single; omissions without fabrication
Sun, 2024 (32)	ChatPDF (GPT-3.5), Claude 2	ChatPDF: Binary 85.9%, Cont 56.1%; Claude: Binary 83.3%, Cont 47.8%	$\kappa = 0.65–0.96$ test–retest	High failure rate for continuous	Binary ≫ continuous (83–86% vs 48–56%)	ChatPDF slightly > Claude 2; continuous challenging
Lai, 2025 (12)	Moonshot-v1-128 k, Claude 3.5 (LLM-assisted workflow)	LLM-assisted: 97.9%; LLM-only: 95.1–96.2%; conventional: 95.3%	Inter-model agreement 93.8%; PABAK = 0.88	62.1% errors from missing info	Variable by domain; Sequence generation: 87.9%	LLM-assisted > LLM-only; 83.1% faster than conventional extraction
Schroeder, 2025 (21)	Gemini 1.5 Pro, Gemini 1.5 Flash, Mistral Large 2	Pro: 72.1–75.8%; Flash: 71.2–73.4%; Mistral: 62.0–62.4%	$\kappa = 0.19–0.34$ (slight-fair)	Mistral: 9/112 failures; 3 incomplete	Explicit: 81–85%; Derived: 65–70%	Gemini Pro ≈ Flash > Mistral; ~25–28% error rates
Ateia, 2025 (23)	Qwen 2.5 72B, Llama 3.3 70B, Gemini 1.5 Flash	Overall score: Llama 3.3 0.548; Qwen 2.5 0.540; Gemini 1.5 Flash 0.490	NR	NR	BERT F1 ~ 0.90 (semantic); Binary: moderate	Llama 3.3 highest overall; open-source competitive with Gemini
Wang,2024 (22)	GPT-4 (gpt-4-turbo-2024–04-09)	Most elements: 90–100%	High for structured	NR	Participant data: excellent; Intervention: similar to humans	Web-based tool; comparable to human coders
Wang, 2025 (33)	GPT-4o, GPT-4, GPT-3.5	Dataset I: 87.9%; Dataset II: 89.0%	Cohen's $\kappa > 0.7$	NR	Improved with standardization	Interactive standardization helps; still < 90%
Andersen, 2025 (34)	Elicit, ChatGPT GPT-4o	Elicit: P = 92.2%, R = 92.2%, F1 = 92.2%; ChatGPT: P = 90.9%, R = 88.8%, F1 = 89.9%	NR	Confabulations: Elicit 4%, ChatGPT 3%	Population characteristics: Elicit 100%, ChatGPT 97%; review-specific variables: 77% vs 80%	Elicit slightly > ChatGPT; AI as second reviewer

Abbreviations: κ, Cohen's kappa; CI, confidence interval; P, precision; R, recall; F1, F1 score; NR, not reported; OR, odds ratio; Cont, continuous; ICC, intraclass correlation coefficient.

percentage points; OR = 1.70, $p < 0.05$). Multiple-sentence prompting improved performance for both models. Caponio et al. (2025) [14] found no significant differences between ChatGPT-4o, Claude 3.5, DeepSeek v3 R1, and Gemini 2.0-flash for dental *meta*-analysis data, though omission patterns varied.

Kataoka et al. (2025) [25] demonstrated that OpenAI's o3 reasoning model outperformed GPT-4o (75.3% vs. 61.6% accuracy on the temporal validation dataset), with the batch extraction method achieving 96.3% mean accuracy on external validation. Schroeder et al. (2025) [21] compared Gemini 1.5 Pro (72.1–75.8%), Gemini 1.5 Flash (71.2–73.4%), and Mistral Large 2 (62.0–62.4%), finding the Gemini models superior but all showing error rates of 25–28%. Khan et al. (2025) [3] evaluated a collaborative two-LLM approach (GPT-4-turbo + Claude-3-Opus with cross-critique), achieving 94% accuracy for the 87% of responses on which the two models agreed, compared with 89% for GPT-4-turbo and 90% for Claude-3-Opus individually, suggesting benefits of multi-model architectures.

Error types and failure modes

Studies characterized several distinct error types. Gartlehner et al. (2025) [11] provided detailed error taxonomy: missed data (5.5%), misallocated data (1.8%), and hallucinated data (0.8%). Importantly, Caponio et al. (2025) [14] found hallucinations were rare (0.08%, 3 out of 3,623 sub-outcomes) while omissions were the dominant error type (60.9–74.5% depending on model and extraction level).

Hallucination rates varied from 0.08% [14] to 6% [13], for categorical variables. Long et al. (2025) [31] found only 1.5% AI inaccuracies versus 18.3% interpretive differences, suggesting most discordance reflects legitimate variation rather than fabrication. Stewart-Evans et al. (2025) [30] reported 1–2 hallucinated findings per 10 sources for complex subjective data. Jansen et al. (2025) [13] found hallucination rates of 2–6% for categorical variables but noted 52% of effect size values were omitted entirely, indicating that failure to extract is more common than fabrication.

Efficiency and practical implementation

Table 4 summarizes efficiency outcomes. Notably, most studies did not quantitatively assess time savings, representing a significant evidence gap.

Gartlehner et al. (2025) [11] provided the most comprehensive efficiency analysis, reporting median time savings of 41 min per study (84 min AI-assisted vs. 125 min human-only), representing a 33% reduction. Lai et al. (2025) [12] compared three approaches: LLM-only, LLM-assisted, and manual extraction, with LLM-assisted extraction achieving the best accuracy-efficiency balance and reducing extraction time by 83.1% (14.7 versus 86.9 min per RCT). Jensen et al. (2025) [18] reported approximately 87% time savings. Gartlehner et al. (2024) [17] did not directly measure time savings, but benchmarked against reported human extraction times of 172 min per study for dual extraction and 107 min for single extraction with verification.

Cost data were rarely reported. Kataoka et al. (2025) [25] did not report costs, noting only that GPT-4o was expensive. Caponio et al. (2025) [14] reported $2.22 for GPT-4o vs $0.22 for DeepSeek v3 R1 (with Claude 3.5 and Gemini available on free tiers). Schmidt et al. (2024) [7] did not report costs. Khan et al. (2025) [3] reported $3.40 per publication for extraction plus $2.06 per publication for cross-critique, and Helms Andersen et al. (2025) [34] reported subscription costs of $12 per month (Elicit) and $20 per month (ChatGPT Plus). Jansen et al.

Table 4
Efficiency Outcomes and Practical Implementation.

Study	Time Savings vs Manual	Cost Estimates	Human Oversight Requirements	Key Limitations	Practical Recommendations
Peng, 2025 (16)	Not assessed	NR	Collaborative approach recommended	No efficiency metrics; terminology challenges	AI as 'second reviewer'
Caponio, 2025 (14)	Not assessed	GPT-4o: $2.22; DeepSeek: $0.22; Claude/Gemini: Free	Human verification essential	High omission rates (60–74%)	Hybrid human-LLM workflows
Ronkainen, 2025 (24)	Not quantified	~€12.60 initial credit	Extensive validation needed	Low recall (0.66); only 41/122 gold-standard results retrieved; single domain	Fine-tuning recommended
Liu, 2025 (19)	~10-13x faster (88.8 s vs 15–20 min)	NR	Targeted review for participant numbers	Baseline characteristics 77.97%; small sample	Use as assistant, not replacement
Daraqel, 2025 (15)	Not quantified	NR	Manual verification required	4.96% discrepancy; complex variables	Supplement to manual extraction
Gartlehner, 2025 (11)	41 min/study saved (33%)	NR	Human review retained	Variable by review (−25 to + 78 min)	Integrate into dual-extraction workflow
Kataoka, 2025 (25)	>80 h saved (estimated)	Not reported (GPT-4o noted as costly)	Required for verification	~60–75% numeric accuracy; hallucinations	Fully automated approach feasible
Khraisha, 2024 (6)	Not quantified	NR	Prompt validation critical	Character limits; inconsistent responses	Substantial caution advised
Jensen, 2025 (18)	~87% time savings	NR	Human review recommended	Hallucinations when info unreported	Use as second rater
Khan, 2025 (3)	Not quantified	$3.40/publication + $2.06 cross-critique	Third reviewer for discrepancies	10 trials only; domain-specific	Two-LLM collaborative approach
Mahmoudi, 2025 (26)	Not quantified	NR	Iterative refinement emphasized	Potential overfitting; single topic	Pilot phase with evaluation
Jansen, 2025 (13)	<6 min per 100 studies	$1.55-$127/100 studies	Essential for effect sizes	Effect sizes: 7–36% accuracy; 52% omission	Suitable for descriptive data only
Gartlehner, 2024 (17)	Not directly measured	Free (Claude 2 browser)	Iterative conversation needed	10 studies; single domain	Promising for initial extraction
Felizardo, 2024 (20)	97% reduction in execution time (JSON optimisation)	95.7% cost reduction (JSON single-iteration)	Author verification performed	Limited to SMS; single domain	Useful for SMS extraction
Bianchi, 2025 (27)	Not quantified	NR	Human comparison essential	Only 20.7% of extractions fully equal to human extraction	Best for simple fields
Sercombe, 2025 (28)	~22 min saved/study (36 min human vs ~ 13.4 min AI-assisted)	NR	Interactive validation needed	89% best still below threshold	Custom GPTs with domain prompts
Meliante, 2025 (29)	Each study < 30 min (Jenni AI)	NR	Extensive oversight required	No single tool best for all tasks	Select tool based on needs
Stewart-Evans, 2025 (30)	Not quantified	NR	Extensive validation needed	Complex data: 10–16% accuracy	Best for simple citation items
Schmidt, 2024 (7)	Not quantified	Not reported	Essential for QA	31% response variation; prompt sensitivity	Feasible for initial extraction
Long, 2025 (31)	Not quantified	NR	Human verification maintained	Interpretive differences: 18.3%	Multi-agent improves accuracy
Sun, 2024 (32)	Not quantified	NR	Manual verification required	Continuous: ~50% accuracy	Binary more reliable than continuous
Lai, 2025 (12)	LLM-assisted: 83.1% faster (14.7 vs 86.9 min/RCT)	NR	Single reviewer sufficient	Chinese OCR challenges; RoB excluded	LLM-assisted hybrid optimal
Schroeder, 2025 (21)	Not quantified	Free API access	Human-in-the-loop essential (AIDE tool)	~25–28% error rates; single domain	HIL workflow with AIDE software
Ateia, 2025 (23)	Not quantified	NR	Expert annotation for validation	Low exact-match accuracy (0.17–0.25); expert annotation burden	API access and programming required
Wang,2024 (22)	Claimed time savings	NR	Validation against gold standard	Small sample (32 studies)	Web-based tool for accessibility
Wang, 2025 (33)	38.9–51.7% reduction vs manual	NR	Interactive standardization helped	40 papers; two domains; <90% accuracy	Interactive interfaces recommended
Andersen, 2025 (34)	Not quantified	$12–20/month subscription	Human review essential	Variable (65.5–88.4%); PDF limitations	AI as second reviewers

Abbreviations: NR, not reported; QA, quality assurance; RoB, risk of bias; OCR, optical character recognition; var, variable; HIL, human-in-the-loop.

(2025) [13] calculated total costs of $2,766.52 for extracting 186 variables across 2,179 PDFs.

All studies emphasized continued need for human oversight. Recommendations ranged from using LLMs as "second reviewers" [16,18,34] to integration into existing dual-extraction workflows [11]. Schroeder et al. (2025) [21] developed the AIDE software tool specifically to facilitate human-in-the-loop validation. No study recommended fully autonomous extraction.

Synthesising across studies, the wide performance range was explained more by methodological differences between studies than by the choice of model. Four study-level features tracked most closely with reported accuracy. First, the granularity of the evaluation unit: studies scoring individual data points (e.g., Caponio et al. [14]) reported near-perfect accuracy, whereas those scoring complete study-level records or derived effect sizes (e.g., Jansen et al. [13]) reported the lowest. Second, the quality of the reference standard: studies using blinded dual human extraction as ground truth (e.g., Gartlehner et al. [11], Kataoka et al. [25]) produced more interpretable estimates than those relying on a single, unverified extractor. Third, the extraction target: descriptive and categorical fields consistently outperformed numerical *meta*-analysis data across heterogeneous domains and models. Fourth, the workflow and prompting design: human-in-the-loop, few-shot, and structured-

output (JSON or codebook) approaches outperformed zero-shot, single-model extraction. Notably, base-model choice (e.g., Claude versus GPT) produced smaller and less consistent differences than these design factors, indicating that how an LLM is deployed and evaluated matters more than which model is used. These methodological axes, rather than the headline accuracy figures alone, provide the most actionable guidance for researchers and underlie the study-level detail tabulated in Tables 1 to 4.

Risk of bias and reporting quality

Methodological quality was assessed using PROBAST + AI, a validated tool for evaluating prediction models incorporating artificial intelligence. Risk of bias was evaluated across four domains: participants and data sources, predictors, outcome, and analysis. Each domain was rated as low, high, or unclear risk. Reporting completeness was assessed using TRIPOD-LLM, a reporting guideline specifically developed for studies using large language models. Two reviewers independently assessed all studies with disagreements resolved through discussion.

Of 27 included studies, 20 (74.1%) were rated as low overall risk of bias, 4 (14.8%) as unclear [6,27,29,and23], and 3 (11.1%) as high risk (Table 5). Domain-specific assessment revealed that participants/data sources (D1) and predictors (D2) showed the lowest risk, with 25/27 studies (92.6%) rated as low risk for D1 and 24/27 (88.9%) for D2. The outcome domain (D3) showed the highest concern, with 3 studies (11.1%) rated as high risk, primarily due to reliance on single human extractors as the reference standard without verification or inter-rater reliability assessment. The analysis domain (D4) identified concerns in 6 studies (22.2%; 3 high and 3 unclear risk) related to inadequate sample sizes (as few as 8 to 25 studies) and lack of test–retest reliability assessment.

Studies rated as high risk of bias included Ronkainen (2025) [24], Mahmoudi (2025) [26], and Felizardo (2024) [20], primarily due to small sample sizes, potential overfitting from using the same data for prompt development and evaluation, and unclear reference standard quality.

Table 5
Risk of Bias Assessment Using PROBAST + AI.

Study	D1	D2	D3	D4	Overall
Peng, 2025 (16)	L	L	L	L	L
Caponio, 2025 (14)	L	L	L	L	L
Ronkainen, 2025 (24)	U	U	H	H	H
Liu, 2025 (19)	L	L	L	L	L
Daraqel, 2025 (15)	L	L	L	L	L
Gartlehner, 2025 (11)	L	L	L	L	L
Kataoka, 2025 (25)	L	L	L	L	L
Khraisha, 2024 (6)	L	L	U	L	U
Jensen, 2025 (18)	L	L	L	L	L
Khan, 2025 (3)	L	L	L	L	L
Mahmoudi, 2025 (26)	U	L	H	H	H
Jansen, 2025 (13)	L	L	L	L	L
Gartlehner, 2024 (17)	L	L	L	L	L
Felizardo, 2024 (20)	L	L	H	H	H
Bianchi, 2025 (27)	L	U	U	U	U
Sercombe, 2025 (28)	L	L	L	L	L
Meliante, 2025 (29)	L	U	U	U	U
Stewart-Evans, 2025 (30)	L	L	U	L	L
Schmidt, 2024 (7)	L	L	L	L	L
Long, 2025 (31)	L	L	L	L	L
Sun, 2024 (32)	L	L	L	L	L
Lai, 2025 (12)	L	L	L	L	L
Schroeder, 2025 (21)	L	L	U	L	L
Ateia, 2025 (23)	L	L	U	U	U
Wang,2024 (22)	L	L	L	L	L
Wang, 2025 (33)	L	L	L	L	L
Andersen, 2025 (34)	L	L	L	L	L

L = Low risk of bias; U = Unclear risk of bias; H = High risk of bias.
Note: Risk-of-bias judgements are reported as text labels in every cell: L (low risk), U (unclear risk) and H (high risk). Across the 27 studies, overall ratings were low in 20 (74.1%), unclear in 4 (14.8%; references 6, 23, 27, 29) and high in 3 (11.1%; references 20, 24, 26).
D1 = Participants/Data Sources; D2 = Predictors; D3 = Outcome; D4 = Analysis.

Reporting quality was generally high, with a mean TRIPOD-LLM compliance score of 88.5% (range: 56–100%). Thirteen studies (48.1%) achieved excellent reporting (≥90%), including Gartlehner (2025) [11], Kataoka (2025) [25], and Schmidt (2024) [7]. Common reporting gaps included incomplete specification of inference settings (temperature, token limits) in 33% of studies, model version and training cutoff dates in 26%, and missing data handling procedures in 30%.

Discussion

This systematic review synthesizes evidence from 27 studies evaluating LLM performance for data extraction in evidence synthesis. Our findings indicate that LLMs achieve variable but often promising accuracy (47–99.9%, with the upper bound reflecting single data-point-level extraction and study-level accuracy typically lower), with performance strongly influenced by data type and complexity. Categorical and string variables are extracted more reliably (74–96%) than numerical data (47–88%), and simple explicitly-stated information outperforms complex derived variables.

Claude models generally demonstrated superior performance to GPT models in direct comparisons [16], though differences were not always statistically significant [14]. The emergence of reasoning-enhanced models (o3) suggests continued improvement. Importantly, omissions appear to be a more common error type than hallucinations, with Caponio et al. (2025) [14] reporting only 0.08% hallucination rate versus 60–74% omission rates. Long et al. (2025) [31] found most discordance (18.3%) reflected interpretive differences rather than AI inaccuracies (1.5%).

Prompting strategy significantly influenced outcomes, with few-shot and iterative approaches outperforming zero-shot prompting. Structured output formats (JSON schemas, predefined tables) facilitated more consistent extraction. Time savings of 33% to 87% were demonstrated where assessed, though most studies did not quantify efficiency.

Most included studies (74.1%) demonstrated low overall risk of bias using PROBAST + AI assessment. The outcome domain presented the greatest concern (11.1% high risk), primarily due to reliance on single human extractors without inter-rater reliability assessment—a notable limitation given the importance of valid ground truth for evaluating LLM performance. The analysis domain identified concerns in 22.2% of studies related to small sample sizes and lack of test–retest reliability. While the evidence base is generally sound, caution is warranted when interpreting results from smaller proof-of-concept studies.

Reporting quality was high (mean TRIPOD-LLM compliance: 88.5%), with nearly half of studies achieving excellent reporting (≥90%). However, common gaps included incomplete specification of inference settings (33%), omitted model versions (26%), and inadequate missing data documentation (30%), limiting reproducibility and underscoring the need for standardized reporting frameworks.

Based on current evidence, LLMs should be used as assistive tools rather than autonomous extractors, with human verification remaining essential; the optimal workflow appears to be LLM-assisted extraction with human review, as demonstrated by Lai et al. (2025) [12] and Gartlehner et al. (2025) [11]. Researchers should match tool to task, recognising that LLMs are better suited for categorical/descriptive data than numerical outcomes, with additional verification warranted for *meta*-analyses requiring precise numerical data. Investment in prompt development is worthwhile, as few-shot prompting with representative examples and iterative refinement consistently improved performance, e.g. Schmidt et al. (2024) [7] demonstrated that approximately 20 iterations may be needed for optimal results. Structured output formats such as JSON schemas or predefined tables facilitate consistent extraction and reduce errors. Finally, transparency regarding model version,

prompting strategy, and validation approach is essential for reproducibility.

This review has several limitations. First, rapid LLM evolution means findings may not generalize to newer model versions. Second, substantial heterogeneity precluded *meta*-analysis. Third, publication bias may favor positive findings. Fourth, many studies used small samples (10–30 articles) and single domains. Additionally, absence of standardized benchmarks makes cross-study comparison challenging. Cost analyses were particularly sparse. Several studies [21] intentionally used non-optimized prompts to simulate realistic researcher usage, which may underestimate achievable performance.

Several priorities emerge for future research: development of standardized benchmarks with diverse tasks and public test sets; consensus reporting guidelines; and large-scale prospective studies comparing LLM-assisted versus traditional workflows. Technical advances including fine-tuning, retrieval-augmented generation, and multi-modal models represent promising directions. The collaborative multi-LLM approach [3] and multi-agent systems [31] merit further investigation. Tools like AIDE [21] that facilitate human-in-the-loop validation should be further developed and validated.

Conclusion

Large language models demonstrate promising but variable performance for data extraction in evidence synthesis. Current evidence supports their use as assistive tools requiring human verification rather than autonomous extractors. Performance varies substantially by data complexity, with categorical variables extracted more reliably than numerical outcomes. Omissions are typically more common than hallucinations, and most discordance reflects interpretive differences rather than fabrication. Where formally measured, time savings of up to approximately 85% were achievable with appropriate implementation, although most included studies did not quantitatively assess efficiency. Future research should prioritize standardized benchmarks, reporting guidelines, and prospective comparative studies.

CRediT authorship contribution statement

Ravi Shankar: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amaevia Lim:** Writing – review & editing, Validation, Formal analysis, Data curation. **Xu Qian:** .

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Data availability

Data will be made available on request.

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